

Using PCA Machine Learning Approach Based on Psychological Questionnaires and Spectral Characteristics of the EEG to Separate the Healthy Participants and Participants with Major Depressive Disorder

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Abstract—Statistical method Principal Component Analysis (PCA) based on psychological questionnaires and EEG spectral characteristics was used to reduce the complex data by finding the most important variables to separate the healthy participants and participants with major depressive disorder from clinic of the State Research Institute of Neurosciences and Medicine. The basic idea of PCA is to identify data patterns that are difficult to see otherwise. In this article it was used PCA for disease classification. We applied PCA to reduce the data dimensionality, for uncovering hidden patterns or relationships between the features. Finally, we trained the classification model on the reduced data to distinguish between healthy participants and participants with major depressive disorder. The study aimed to investigate the relationship between psychological factors and EEG spectral characteristics in participants with major depressive disorder. The study found that there was a significant correlation between the severity of depression and the EEG spectral characteristics. These findings suggest that the combination of psychological questionnaires and EEG spectral characteristics could be a useful tool for the diagnosis and classification of major depressive disorder.

Keywords—machine learning, PCA, hypothesis testing, electroencephalogram, processing of experimental data

I. INTRODUCTION

Neurophysiology is the branch of physiology that studies the function and properties of neurons, including their interactions with each other and with other cells in the body, as well as the mechanisms that underlie sensory and motor functions, cognition, and behavior. It involves the study of electrical and chemical signaling in the nervous system, and the use of techniques such as electrophysiology, neuroimaging, and molecular biology to understand the complex workings of the brain and nervous system.

Neurophysiology is an important area of research in both basic and clinical neuroscience. Basic research in neurophysiology has helped us to better understand the fundamental mechanisms that govern the functioning of the nervous system, including how neurons communicate with one another, how sensory information is processed, and how motor actions are coordinated. This knowledge has led to the development of new therapies for neurological and psychiatric disorders.

Clinical neurophysiology, on the other hand, is concerned with the diagnosis and treatment of neurological disorders, such as epilepsy, multiple sclerosis, and Parkinson's disease, using techniques such as electroencephalography (EEG), electromyography (EMG), and nerve conduction studies. These techniques allow clinicians to measure the electrical activity of the brain, nerves, and muscles, and to use this information to diagnose and monitor various neurological conditions. One of the notable benefits of electroencephalography (EEG) methods is the capacity to directly record neuronal activity [1]. This is achieved by quantifying voltage fluctuations on the surface of the head that arise from ion current flows in brain neurons.

The scientific significance and novelty of the study's findings lie in their identification of the involvement of brain structures, particularly in relation to their electrical activity, in the provision of reactive social behavior. Furthermore, the study contributes to the understanding of the role played by the individual in these processes. The activity of brain regions such as the medial prefrontal cortex and the right temporal parietal cortex, which are recognized as the main centers of the "social brain," is instrumental in mediating reactive social behavior. In addition, it is postulated that the functioning of these mechanisms may be linked to personal characteristics,

specifically to the dimension of "friendliness" (agreeableness) as outlined in the five-factor personality model [2]. Simultaneously, there exist established methods, such as the sLORETA software package [3], which are capable of determining the sources of electrical signals within the brain by analyzing EEG signals obtained from the head surface. Consequently, EEG methods may be effectively utilized in the study of high-speed brain processes [4]. Differential Psychophysiology at the State-Research Institute of Neurosciences and Medicine has actively conducted research, including the collection of EEG data. However, the development of proprietary software to automate this process is an extremely time-consuming task [5]. Consequently, a critical issue remains the identification of a scheme for processing heterogeneous EEG data that relies on existing software components.

Overall, the study of neurophysiology is critical to our understanding of how the nervous system functions and how it is affected by disease and injury, and has important implications for the development of new therapies and treatments for a wide range of neurological and psychiatric conditions. It was found that depression is one of the most distributed in the world affective disorders, characterized by complex behavioral and neurophysiological symptoms [6]. Early diagnostics of depression is very important that is why there are many methods to prevent the consequences of this disease [6, 7].

The present article outlines an automated methodology for testing hypotheses utilizing diverse EEG data sources, which includes the use of software and tools for collecting experimental data, as well as developed algorithms for their preliminary processing and standardization. We applied PCA to reduce the dimensionality of the data, for uncovering hidden patterns or relationships between the features. Finally, we trained the classification model on the reduced data to distinguish between healthy participants and participants with major depressive disorder.

II. OBTAINING EGG DATA

Statistical techniques are utilized for hypothesis testing based on data gathered through the monitoring of study participants. The input data encompasses the results of subject questioning pertaining to variables such as gender, age, social status, nationality, among others, as well as computer recordings obtained during the course of the experiment.

During the EEG recording, a specialized helmet is placed on the participant's head, with electrodes connected to an electroencephalograph (Fig. 1). ~~Participants were from the clinic of the State Research Institute of Neurosciences and Medicine.~~

The electrical potential at each electrode is measured and recorded at a sampling rate of 1 kHz. The potential of the electrode located at the top of the head serves as the reference or zero potential. During the experiment, the subject is presented with elementary test tasks referred to as epochs. The tasks begin by displaying a stimulating picture, such as a photograph of a human face expressing an emotional state, and a text with reaction options or a list of possible actions on a monitor positioned in front of the subject. The subject selects an option using a keyboard, after which the next epoch begins.

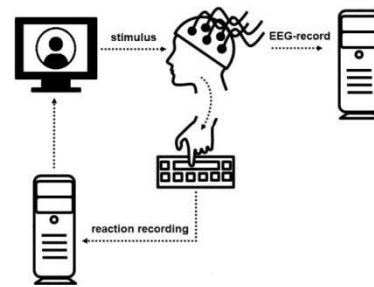


Fig. 1. Experimental data analysis design.

The experiment is conducted automatically, where the stimuli presentation and reaction recording is controlled by the INQUISIT program [8]. Data is recorded using either Neuroscan (EEG cap with 118 electrodes) or Brain Products (EEG cap with 128 electrodes) electroencephalographs, along with their respective proprietary software.

Following the experiment, the coordinates of the electrodes and reference points of the EEG helmet are measured using the FASTRAK digitizer (Polhemus) [9]. This measurement serves to characterize the shape of the subject's head. The location of the reference points is established by employing a standardized procedure, which includes identifying specific landmarks on the ears, nose, and back of the head.

The EEG was recorded using a 128-channel Brain Products amplifier, Germany, bandwidth 0.1-100 Hz, signal sampling frequency 1000 Hz, with a reference electrode Cz. The artifacts were removed with the independent components analyses by means of eeglab_toolbox [10]. Spectral power values were calculated separately for each subject and for each EEG channel, in the ranges delta (1-4 Hz), theta (4-8 Hz), alpha1 (8-10 Hz), alpha2 (10-12 Hz), beta1 (12-16 Hz), beta2 (16-20 Hz), beta3 (20-25 Hz) and gamma (25-35 Hz) rhythms. After that, the relative spectral power index was calculated (spectral power for one rhythm divided by the spectral power in the range from 1 to 35 Hz and multiplied by 100). It was made separately for each subject, EEG channel and EEG rhythm.

These indices were averaged for each participant from the 11 EEG electrodes groups: left frontal region, middle frontal region, right frontal region, left temporal region, central region, right temporal region, left parietal-occipital region, middle parietal-occipital region, right parietal region, occipital region. Relative spectral power indices for each cortical region in each participant separately for EEG recordings with open and closed eyes were used in conjunction with assessments on different scales of psychological tests to train the PCA model.

III. EEG DATA PROCESING AND UNIFICATION

In this article, the PCA machine learning method was used for data processing to divide healthy and unhealthy participants into classes. Principal Component Analysis (PCA) is a technique used in data analysis to identify patterns in high-dimensional data by reducing its dimensionality while retaining most of its variability. It is a linear transformation that identifies the axes of greatest variance in the data and projects the data onto those axes, producing a new set of variables called principal components. PCA is commonly used in fields such as image processing, genetics,

and finance to extract meaningful information from large datasets [11].

PCA works by finding the eigenvectors and eigenvalues of the covariance matrix of the data. The eigenvectors represent the directions in the data with the highest variance, while the eigenvalues represent the amount of variance explained by each eigenvector. The eigenvectors can be used to transform the data into a new coordinate system, where the first principal component explains the largest possible amount of variance, and each subsequent component explains as much of the remaining variance as possible.

PCA can be used for various tasks, such as data compression, feature selection, and data visualization. It is a powerful tool for identifying patterns and relationships in high-dimensional data, and it can help to reduce the noise and redundancy in the data, making it easier to interpret and analyze. To perform PCA, the following steps are typically followed:

- Standardize the data: This involves scaling the data so that each variable has a mean of 0 and a standard deviation of 1.
- Compute the covariance matrix: This involves calculating the covariance between each pair of variables in the data.
- Compute the eigenvectors and eigenvalues of the covariance matrix: This involves finding the directions of greatest variance in the data and the amount of variance explained by each direction.
- Choose the number of principal components: This involves deciding how many principal components to retain based on the amount of variance they explain and the desired level of dimensionality reduction.
- Transform the data: This involves projecting the original data onto the chosen principal components to create a new dataset in the reduced dimensional space.

PCA can be implemented in various programming languages such as Python, R, MATLAB, and others, using built-in functions or libraries. It is a widely used technique in various fields, including machine learning, data science, and statistics, and it can provide valuable insights and simplify the analysis of high-dimensional data. In our work we used Python language.

Processing of psychometric data.

In our work 71 people were selected as a database, of which 42 were participants with healthy characteristics. We included 29 patients who met criteria for current MDD. Patients were selected from The Patient Data Base (permission and property of the Department of Clinical Neuroscience, Behavior and Neurotechnology, State-Research Institute of Neuroscience and Medicine). Psychometric and spectral characteristics were known for each person from this sample. Psychometric characteristics include the following 22 parameters: gender, age, Beck's questionnaire [12], general tendency to negative emotions, tendency to ponder negative emotions and find positive moments, tendency to suppress bad mood, obsessive thoughts, extroversion, awareness, neuroticism, intelligence, positive expression, negative expression, attention to other people's emotions, enjoyment of emotions,

empathy [13], anger, psychological aggression, hostility, verbal aggression [14]. The total amount of data for the description of psychometric data is (71, 22, 1). Here 71 is the number of rows in the data, 22 is the number of parameters studied.

The obtained electroencephalography data, in particular alpha, beta, gamma and delta rhythms, were used as spectral characteristics. 17 parameters were identified for each of the rhythms, in particular: the left frontal zone with closed eyes (LF_close), the medial frontal zone with closed eyes (MF_close), the right frontal zone with closed eyes (RF_close), the left temporal zone with closed eyes (LT_close), the right parietal zone with open eyes, the central occipital zone with closed eyes, the right occipital zone with open eyes, left frontal zone with open eyes (LF_open), right frontal zone with open eyes (RF_open), left temporal zone with open eyes (LT_open), medial occipital zone with open eyes (MOP_open), right occipital zone with open eyes (ROP_open). The total number of rhythms was 8 and includes data for the rhythm: alpha1, alpha2, beta1, beta2, beta3, delta, gamma and theta.

Thus, the total volume of processed data on spectral characteristics is (71, 17, 8). Here 71 is the number of rows in the data, 17 is the number of parameters studied, 8 is the number of frequencies used.

At the first stage of data processing, the analysis of data on psychometric characteristics was carried out. The following psychometric data were averaged: joy_emp_min, mis_emp_min, pos_expr_min, empathy_min, des_emot_min, att_emot_min, neg_expr_min.

Averaging was carried out as follows: a correlation matrix of the influence of these characteristics on health was built, Fig.2 (a), if the trait correlated negatively with health, then a modification was made for negatively correlating data, which consisted in multiplying these data by (-1), after which an updated correlation matrix was built Fig. 2 (b). After that, the data for the selected psychometric parameters were averaged into a new averaged attribute: Emin_Av.

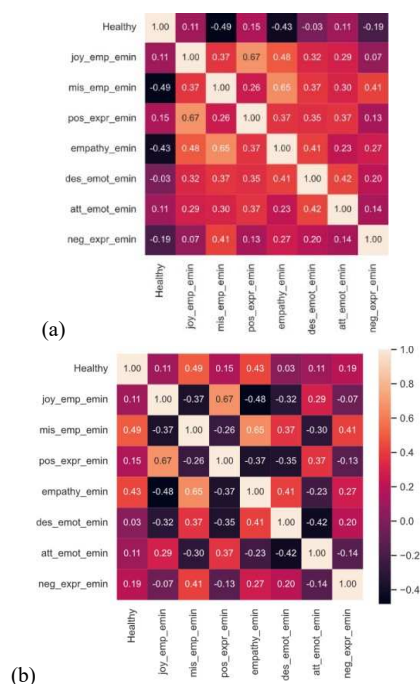


Fig. 2. Averaging of selected psychometric characteristics. (a) the initial

correlation matrix, (b) the correlation matrix after modification.

A similar averaging procedure was performed for psychometric data: hostel, physagr, anger, vebagr. The correlation matrices of these characteristics on health before and after modification are shown in Fig. 3 (a, b). After that, the data for the selected psychometric parameters were averaged into a new averaged trait: Agr_Av.

Thus, the description of psychometric data is reduced from 22 parameters to $d=10$: gender, age, bdi_sum, Extr, Cons, Neur, Intel, Emin_Av, Agr_Av, Emot_Av. The next step in the processing of psychometric data is the application of the linear transformation method PCA to identify the most significant parameters. Since we want to lower the dimension of the dataset by compressing it into a new feature subspace k , we select only a subset of the eigenvectors (principal components) that contain most of the information (variance).

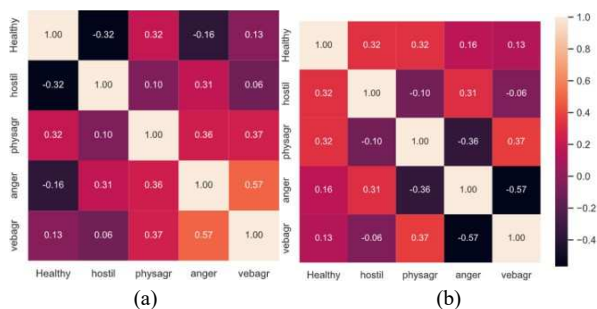


Fig. 3. Averaging of selected psychometric characteristics. (a) the initial correlation matrix, (b) the correlation matrix after modification.

The eigenvalues determine the magnitude of the eigenvectors. In order to extract the most relevant information from the eigenvectors, it is necessary to sort the corresponding eigenvalues in descending order of magnitude. The magnitude of the eigenvalues determines the significance of the corresponding eigenvectors, with larger magnitudes indicating greater importance. Therefore, we focus on the upper k eigenvectors that correspond to the eigenvalues with the highest magnitudes. To examine the variance explained ratio of the eigenvalues, we can plot their coefficients.

The coefficient of the explained variance of the eigenvalue λ_i is the fraction of the eigenvalue λ_i in the total sum of the eigenvalues: $\lambda_i / \sum^d \lambda_i$.

To find out the degree of influence of features using the PCA method, we project the data of 10 initial features onto a two-dimensional space, this will allow us to visually represent the values of the eigenvalues and then sort them in descending order of their module. Figure 6 (a) shows the values of the eigenvalue module for normalized psychometric data gender, age, bdi_sum, Extr, Cons, Neur, Intel, Emin_Av, Agr_Av, Emot_Av (total $d=10$) in two-dimensional space.

Processing of spectral data.

The obtained alpha, beta, gamma and delta rhythm data were used as spectral characteristics. The total number of rhythms was 8 and includes data for the rhythm: alpha1, alpha2, beta1, beta2, beta3, delta, gamma and theta. The following 17 parameters were used for each of the rhythms: LF_close, MF_close, RF_close, LT_close, MC_close, RT_close, LOP_close, MOP_close, ROP_close, LF_open, RF_open, LT_open, MC_open, RT_open, LOP_open, MOP_open, ROP_open.

The dimensionality of the data was reduced by the averaging method, taking into account the correlation of the values among themselves.

- The parameters LOP_open, MOP_open, ROP_open are averaged into the general parameter OP_Av_open
 - The parameters LOP_close, MOP_close, ROP_close are averaged into the general parameter OP_Av_close
 - The parameters LF_open, RF_open are averaged into the general parameter F_Av_open
 - The parameters LF_close, MF_close, RF_close are averaged into the general parameter F_Av_close
 - LT_open, RT_open parameters are averaged into the general parameter T_Av_open
 - LT_close, RT_close parameters are averaged into the general parameter T_Av_close
- Thus, the description of the spectral parameters for each rhythm is reduced from 17 parameters to 8: MC_close,

Then we can calculate the cumulative sum of the explained variances and display it on a graph. The resulting graph (Fig. 4) shows that the share of the 6 main components accounts for more than 95% of the variance in the data set.

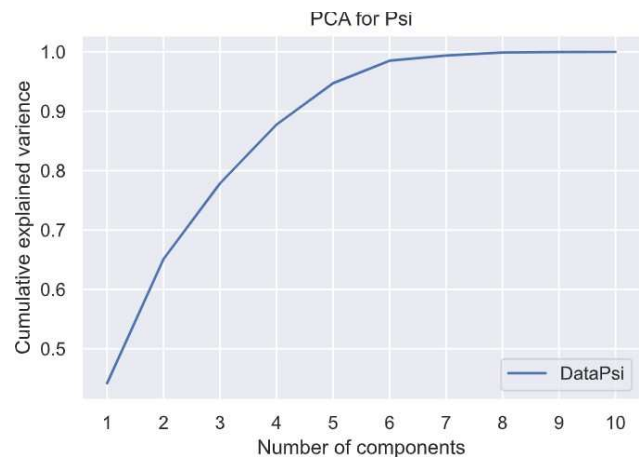


Fig. 4. Graph of the explained variance for psychometric data.

MC_open, OP_Av_open, OP_Av_close, F_Av_close, F_Av_open, T_Av_close, T_Av_open. The resulting graph of explained variance for all frequencies, Fig. 5.

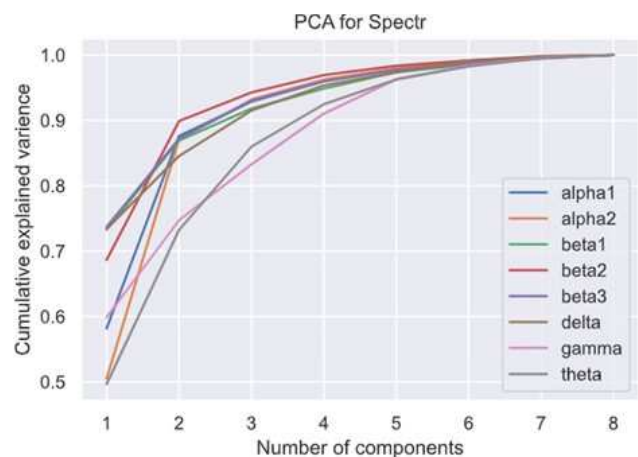


Fig. 5. Graph of the explained variance for spectral data.

IV. RESULTS

The obtained data for each of the frequencies were projected by the PCA method onto a two-dimensional space, which made it possible to visually imagine the possibility of dividing the sick and healthy into separate classes, Fig 6 (a), shows the representation of participants with major depressive disorder and healthy participants in two-dimensional space for the frequencies alpha1 and alpha2.

The line between classes is drawn using the Support VectorMachine (SVM) method. The use of this algorithm allowed to minimize errors of incorrect classification. In particular, Fig. 6 (a) shows that the boundary of solutions between the two classes can be drawn in various ways.

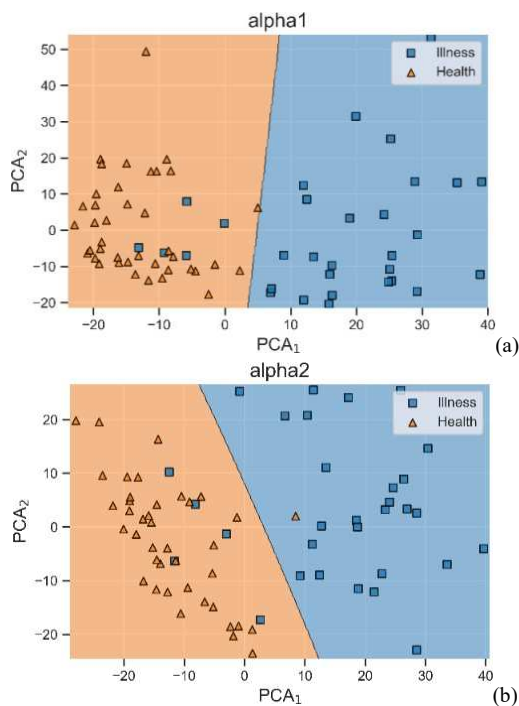


Fig. 6. Representation of the healthy participants and participants with major depressive disorder in two-dimensional space for alpha frequencies.

In the SVM method, the main goal of optimization is to determine the maximum distance between the separating hyperplane and the training samples closest to this hyperplane, which are called support vectors, Fig. 6 (b).

Fig. 7 (a, b, c) shows the representation of participants and healthy in two-dimensional space for the frequencies beta1, beta2 and beta3. The line between the classes is drawn using the Support Vector Machine (SVM).

Fig. 8 (a, b, c) shows the representation of participants with major depressive disorder and healthy participants in two-dimensional space for the frequencies delta, gamma and theta. The line between the classes is drawn using the Support vector Machine (SVM), [15, 16]. A modified kernel Support vector Machine method (kernel SVM) was used to solve the non-linear classification problems. One of the most common kernels is the Radial Basic Function (RBF) kernel or Gaussian kernel.

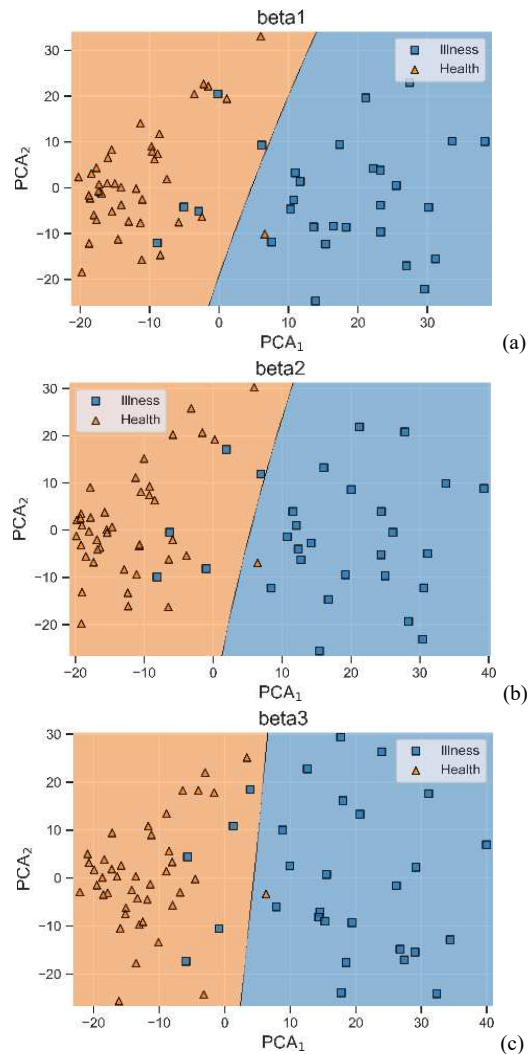
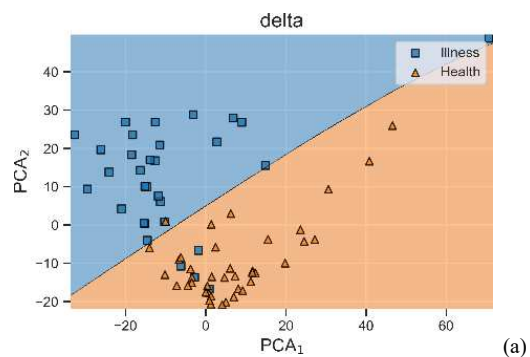


Fig. 7. Representation of the healthy participants and participants with major depressive disorder in two-dimensional space for beta frequencies.

RBF kernel SVM with parameter $\gamma=0.0001$ was used to distinguish between healthy and sick classes. The smaller γ , the greater the variance of the allocated classes. The optimal parameter γ was established by numerical experiments and its small value indicates that the boundary between classes can be drawn with good accuracy by linear classification.



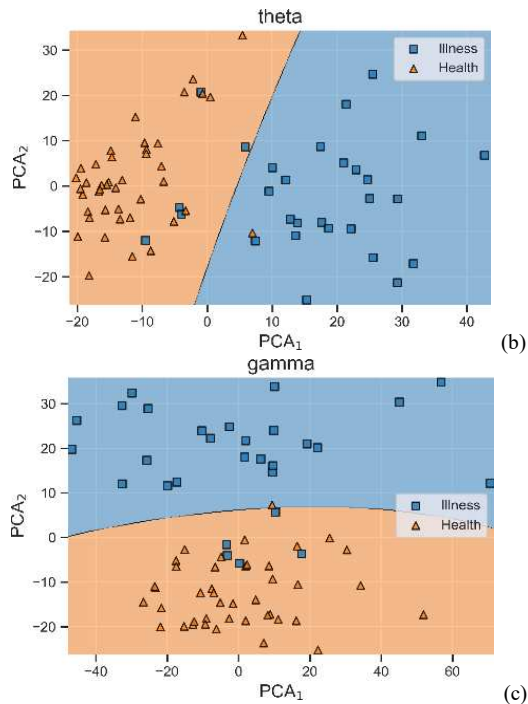


Fig. 8. Representation of the healthy participants and participants with major depressive disorder in two-dimensional space for delta, gamma, theta frequencies.

V. CONCLUSION

On the basis of the data selected from the databases, we formed a data array including psychometric and spectral characteristics of the subjects, focused on the research and development of algorithms for classifying (diagnosing) the subjects using machine learning methods. Using the method of averaging such characteristics, the adapted PCA algorithm and the “Varimax rotation” method, we identified insignificant parameters and made a fourfold reduction in the dimension of the original dataset. Using the Support Vector Machine with a hard gap (hard-margin SVM), we determined a hyperplane separating the dataset, which shows at least 97% reliability of dividing the subjects diagnosed with major depressive disorder and healthy control subjects into separate classes.

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